Database Schema Design - British Auction in RFQ System

Entity-Relationship (ER) Diagram & Relational Data Dictionary

Database Engine: SQLite 3 Relational Database	**ORM:** SQLAlchemy 2.0 (Declarative Base)
Normalization: Third Normal Form (3NF)	**Integrity:** Foreign Key Constraints & Cascade Delete
Total Tables: 7 Core Normalized Entities	**File Location:** backend/auction.db

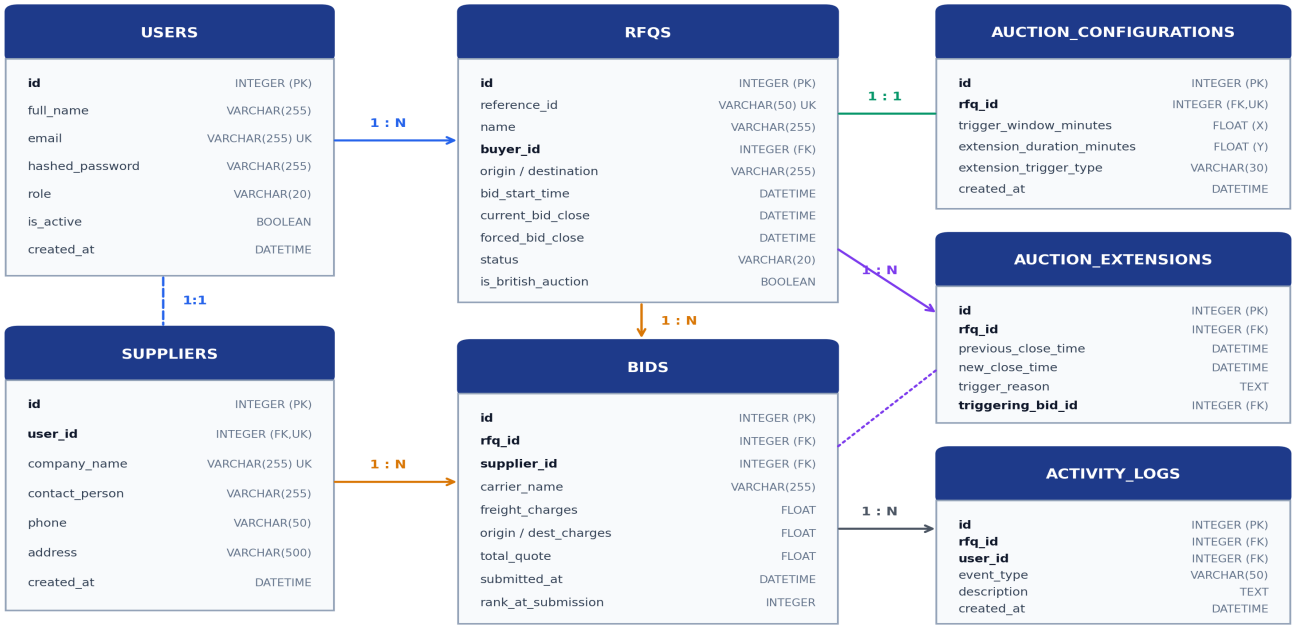
1. Database Architecture & Design Principles

The British Auction RFQ database schema is structured into **7 normalized relational tables** designed to decouple user authentication, supplier commercial profiles, procurement RFQ specifications, British auction configuration parameters, quote transactions, dynamic time extensions, and audit trails.

Core Schema Design Pillars:

- **Referential Integrity:** Enforced using foreign key constraints with cascading deletion rules (e.g. deleting an RFQ removes its configuration, quote records, and extension logs cleanly).
- **High-Throughput Indexing:** Unique and B-Tree indexes placed on lookup keys: email, reference_id, rfq_id, supplier_id, submitted_at, and total_quote.
- **Complete Auditability:** Dedicated activity_logs table records every administrative, bidding, and extension action as an immutable chronological ledger.

2. Entity-Relationship (ER) Diagram



3. Detailed Data Dictionary & Table Definitions

Table: users — User credentials, access roles, and platform status

Column Name	Data Type	Constraints & Keys	Description & Purpose
`id`	INTEGER	PRIMARY KEY, AUTO	Unique identifier for user account
`full_name`	VARCHAR(255)	NOT NULL	Full legal name of user
`email`	VARCHAR(255)	NOT NULL, UNIQUE, INDEX	Work email address used for login
`hashed_password`	VARCHAR(255)	NOT NULL	Salted bcrypt hash of user password
`role`	VARCHAR(20)	NOT NULL	System role: ADMIN, BUYER, or SUPPLIER
`is_active`	BOOLEAN	NOT NULL, DEFAULT TRUE	Active account toggle
`created_at`	DATETIME	NOT NULL, DEFAULT UTC	Account creation timestamp

Table: suppliers — Company details for registered logistics suppliers and carrier operators

Column Name	Data Type	Constraints & Keys	Description & Purpose
`id`	INTEGER	PRIMARY KEY, AUTO	Unique supplier entity identifier
`user_id`	INTEGER	**NOT NULL, UNIQUE, FK -> users.id**	Reference to parent user account
`company_name`	VARCHAR(255)	NOT NULL, UNIQUE, INDEX	Registered business name (e.g. Maersk)
`contact_person`	VARCHAR(255)	NOT NULL	Primary representative contact name
`phone`	VARCHAR(50)	NULLABLE	Contact telephone number
`address`	VARCHAR(500)	NULLABLE	Registered corporate business address
`created_at`	DATETIME	NOT NULL, DEFAULT UTC	Supplier registration timestamp

Table: rfqs — Request for Quotations (RFQs) published by buyers

Column Name	Data Type	Constraints & Keys	Description & Purpose
`id`	INTEGER	PRIMARY KEY, AUTO	Unique RFQ record identifier
`reference_id`	VARCHAR(50)	NOT NULL, UNIQUE, INDEX	Human-readable identifier (e.g. GC-RFQ-2026-001)
`name`	VARCHAR(255)	NOT NULL, INDEX	Procurement title or freight route name
`description`	TEXT	NULLABLE	Detailed shipping cargo requirements
`buyer_id`	INTEGER	**NOT NULL, FK -> users.id**	Reference to buyer user creating the RFQ
`origin / destination`	VARCHAR(255)	NOT NULL	Origin and Destination seaport/airport locations
`cargo_details`	VARCHAR(255)	NULLABLE	Cargo volume, weight, container type
`pickup_service_date`	DATETIME	NOT NULL	Expected cargo pickup date
`bid_start_time`	DATETIME	NOT NULL	Scheduled start time when bidding opens
`original_bid_close_time`	DATETIME	NOT NULL	Original scheduled closing time
`current_bid_close_time`	DATETIME	NOT NULL, INDEX	Effective closing time (dynamically extended)
`forced_bid_close_time`	DATETIME	NOT NULL	Absolute maximum forced close ceiling
`status`	VARCHAR(20)	NOT NULL	Status: SCHEDULED, ACTIVE, CLOSED, FORCE_CLOSED
`is_british_auction`	BOOLEAN	NOT NULL, DEFAULT TRUE	Enables British Auction anti-sniping rules
`target_price`	FLOAT	NULLABLE	Internal baseline benchmark price
`created_at / updated_at`	DATETIME	NOT NULL, DEFAULT UTC	Audit timestamps for record creation/update

3. Detailed Data Dictionary (Continued)

Table: auction_configurations — British Auction parameters for each RFQ

Column Name	Data Type	Constraints & Keys	Description & Purpose
id	INTEGER	PRIMARY KEY, AUTO	Unique configuration identifier
rfq_id	INTEGER	**NOT NULL, UNIQUE, FK -> rfqs.id**	Parent RFQ reference (1-to-1 relationship)
trigger_window_minutes	FLOAT	NOT NULL, DEFAULT 10.0	X minutes: Anti-sniping monitoring window
extension_duration_minutes	FLOAT	NOT NULL, DEFAULT 5.0	Y minutes: Time added per qualifying trigger
extension_trigger_type	VARCHAR(30)	NOT NULL	Trigger condition: ANY_BID, ANY_RANK_CHANGE, or L1_RANK_CHANGE
created_at	DATETIME	NOT NULL, DEFAULT UTC	Configuration timestamp

Table: auction_extensions — Audit trail of each dynamic time extension event

Column Name	Data Type	Constraints & Keys	Description & Purpose
id	INTEGER	PRIMARY KEY, AUTO	Unique extension record identifier
rfq_id	INTEGER	**NOT NULL, INDEX, FK -> rfqs.id**	Target RFQ reference
previous_close_time	DATETIME	NOT NULL	Closing timestamp prior to this extension
new_close_time	DATETIME	NOT NULL	Extended closing timestamp
extension_duration	FLOAT	NOT NULL	Minutes added by this extension
trigger_reason	TEXT	NOT NULL	Detailed explanation of trigger and supplier
triggering_bid_id	INTEGER	**NULLABLE, FK -> bids.id**	Specific quote that triggered the extension
created_at	DATETIME	NOT NULL, DEFAULT UTC	Timestamp when extension was calculated

Table: bids — Quotes submitted by suppliers with price breakdown and carrier data

Column Name	Data Type	Constraints & Keys	Description & Purpose
id	INTEGER	PRIMARY KEY, AUTO	Unique quote identifier
rfq_id	INTEGER	**NOT NULL, INDEX, FK -> rfqs.id**	Target RFQ reference
supplier_id	INTEGER	**NOT NULL, INDEX, FK -> suppliers.id**	Submitting supplier account reference
carrier_name	VARCHAR(255)	NOT NULL	Operating ocean carrier or airline fleet name
freight_charges	FLOAT	NOT NULL	Base ocean / air freight transportation price
origin / dest_charges	FLOAT	NOT NULL, DEFAULT 0.0	Origin THC and destination discharge port charges
total_quote	FLOAT	NOT NULL, INDEX	Calculated sum: freight + origin + destination
transit_time	INTEGER	NOT NULL	Estimated cargo transit duration in days
validity_of_quote	DATETIME	NOT NULL	Quote rate validity expiration timestamp
submitted_at	DATETIME	NOT NULL, INDEX, DEFAULT UTC	Precise submission timestamp
is_valid	BOOLEAN	NOT NULL, DEFAULT TRUE	Active/valid status flag
rank_at_submission	INTEGER	NULLABLE	Rank achieved at moment of submission

Table: activity_logs — Immutable audit trail of system, RFQ, and auction events

Column Name	Data Type	Constraints & Keys	Description & Purpose
id	INTEGER	PRIMARY KEY, AUTO	Unique audit log identifier
rfq_id	INTEGER	**NULLABLE, INDEX, FK -> rfqs.id**	Associated RFQ (if applicable)
user_id	INTEGER	**NULLABLE, INDEX, FK -> users.id**	User who initiated the action
event_type	VARCHAR(50)	NOT NULL	Event category (RFQ_CREATED, BID_SUBMITTED, etc.)
description	TEXT	NOT NULL	Human-readable event narrative
created_at	DATETIME	NOT NULL, INDEX, DEFAULT UTC	Event creation timestamp